Week 5 Homework: AutoML & No-Code Model Training

Train a Custom Model, Understand Metrics, Compare Generic vs Fine-Tuned

Estimated Time: 2.5–3 hours **Difficulty:** Beginner-Intermediate

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Homework Objectives

By the end of this homework, you will have:

1. A custom-trained image classifier built with Google Teachable Machine
2. A calculated confusion matrix with accuracy, precision, and recall for your model
3. Two fine-tuned Hugging Face models tested on your capstone-relevant data
4. An n8n workflow comparing generic vs fine-tuned model performance
5. A training and evaluation report documenting your findings
6. All deliverables uploaded to your GitHub repository

> **Domain Options**
>
> This lab has two tracks. **Part 1 (Teachable Machine)** uses images — use the image category that fits your domain. **Parts 3–4 (fine-tuned models)** use text — use the text inputs from your domain.

Your Domain	Image Classification (Part 1)	Text Inputs (Parts 3–4)
Cybersecurity	Phishing vs legitimate email screenshots	"Unauthorized login from IP 198.51.100.4 traced to Moscow targeting user John Miller"
Digital Forensics	Suspicious vs normal file browser screenshots	"Deleted file recovery attempted by analyst James Chen on NYPD workstation WS-047"
Fraud Investigation	Forged vs authentic document images	"Wire transfer of $47,500 from Chase Bank account to Lagos — account opened 3 days ago"
Intelligence Analysis	Relevant vs irrelevant map/satellite images	"Source reports meeting between known extremist group and foreign nationals in Zurich"
Forensic Psychology	Distressed vs calm facial expressions	"Dr. Sarah Lopez flagged escalating verbal hostility during intake at Rikers facility"
Criminal Justice	Handwritten vs typed/printed text samples	"Defendant Robert Garcia failed to appear at Manhattan Criminal Court, third occurrence"
Security Management	Authorized vs unauthorized access badge photos	"Badge access by Apex Security contractor at Northfield HQ server room 3B outside hours"

Prerequisites

- Completed Week 1–4 homework (n8n account, Airtable account, Hugging Face token, Groq key, GitHub repo)
- Web browser (Chrome or Firefox)
- Your Week 4 model comparison workflow (you'll build on those results)

No new software installation required!

Part 1: Train a Custom Image Classifier with Teachable Machine (45 minutes)

You're going to train an image classifier from scratch — no code, no installation, no account. This is the fastest way to understand what "model training" actually means.

Step 1.1: Open Teachable Machine

1. Go to: **https://teachablemachine.withgoogle.com**
2. Click **"Get Started"**
3. Choose **"Image Project"**
4. Choose **"Standard image model"**

You'll see the training interface with two default classes ("Class 1" and "Class 2").

Step 1.2: Set Up Your Classes

Rename the classes to match your domain:

For Cybersecurity (default):

- Class 1: `Phishing`
- Class 2: `Legitimate`

For Forensic Psychology:

- Class 1: `Distressed`
- Class 2: `Calm`
- Search Google Images for "distressed facial expression" and "calm facial expression" or "neutral portrait"

For Criminal Justice:

- Class 1: `Handwritten`
- Class 2: `Typed`
- Search Google Images for "handwritten note" and "typed document" or "printed text"

For other domains: Use the image classification categories from the Domain Options table above.

> **Optional:** Click "Add a class" to add a third class (e.g., `Suspicious` or `Uncertain`) for a more realistic multi-class problem.

Step 1.3: Collect Training Images

You need **at least 20 images per class** (30–50 is better). Three ways to collect them:

Option A — Google Image Search (recommended):

1. Search Google Images for "phishing email example" and "phishing email screenshot"
2. Right-click and save 25–30 images to a folder called `phishing/`
3. Search for "email inbox screenshot" and "legitimate email"
4. Save 25–30 images to a folder called `legitimate/`
5. In Teachable Machine, click **"Upload"** on each class and select your images

Option B — Webcam (quick demo):

1. Print or display phishing email examples on your screen
2. Use the webcam option in Teachable Machine to capture images of the screen showing phishing emails
3. Do the same for legitimate emails
4. This is less realistic but works for understanding the training process

Option C — Provided dataset: If your instructor has provided a shared image folder, download and use those images.

> **Important:** Make your data diverse. Don't use 20 screenshots of the same email. Include different email clients, different phishing styles, different legitimate email types. Diversity in training data is what makes models generalize.

Step 1.4: Set Aside Test Images

Before you train, remove 5 images from each class and save them in a separate folder called `test/`. These images will NOT be used for training — you'll use them to evaluate the model honestly.

Your final setup:

- Training: ~20–25 images per class (in Teachable Machine)
- Test: 5 images per class (saved separately on your computer)

Step 1.5: Train the Model

1. Click **"Train Model"**
2. Watch the training progress bar — this typically takes 15–60 seconds
3. When done, the "Preview" panel on the right activates

Step 1.6: Test Your Model

1. In the Preview panel, switch to **"File"** input (instead of Webcam)
2. Upload each of your 10 test images **one at a time**
3. For each image, record the results in this table:

#	File Name	Actual Class	Predicted Class	Confidence
1	test_phish_01.png	Phishing		

#	File Name	Actual Class	Predicted Class	Confidence
2	test_phish_02.png	Phishing		
3	test_phish_03.png	Phishing		
4	test_phish_04.png	Phishing		
5	test_phish_05.png	Phishing		
6	test_legit_01.png	Legitimate		
7	test_legit_02.png	Legitimate		
8	test_legit_03.png	Legitimate		
9	test_legit_04.png	Legitimate		
10	test_legit_05.png	Legitimate		

Step 1.7: Take Screenshots

Take screenshots of:

1. Your Teachable Machine training interface (showing both classes with images loaded)
2. At least 2 test results — one correct prediction and one incorrect or low-confidence prediction
3. Save these in a `screenshots/` folder for your GitHub submission

Step 1.8: Export Your Model (Optional)

1. Click **"Export Model"**
2. Choose **"Tensorflow"** → **"Keras"**
3. Click **"Download my model"**
4. This downloads `converted_keras.zip` containing `keras_model.h5` and `labels.txt`
5. Save these — you'll use them in the bonus challenge if you want to deploy to Hugging Face Spaces

> **Note:** Exporting is optional for this lab. The core deliverable is your test results and metrics analysis, not a deployed model.

Part 1 Alternative: Text Classification with Google Colab (optional)

All domains now have image classification options above, so this section is **optional**. If you'd prefer to train a text classifier instead of an image classifier — or want to try both — follow the steps below. No coding experience is required: you'll copy and paste pre-written cells and just swap in your own text examples.

1. Go to **Google Colab**: https://colab.research.google.com
2. Create a **new notebook**
3. Copy and run these cells one at a time (you don't need to understand the Python — just run each cell and read the output):

Cell 1 — Install and import:

```
!pip install -q scikit-learn pandas
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix
```

Cell 2 — Create your labeled dataset (edit these to match your domain):

```
# Replace these with 20+ examples relevant to YOUR domain
data = {
    'text': [
        # High concern examples (label: "high")
        "Subject made explicit threats toward former partner during intake
interview",
        "Defendant failed to appear for third scheduled court date warrant
issued",
        "Evaluee exhibited significant agitation during structured assessment",
        "Client reported active ideation with specific plan during session",
        "Subject refused to provide current address and denied supervision terms",
        "Parolee tested positive for controlled substance third violation",
        "Inmate involved in altercation resulting in injury to staff member",
        "Client missed four consecutive appointments without notification",
        "Subject displayed escalating hostility and made veiled threats",
        "Defendant violated protective order contacted victim directly",
        # Low concern examples (label: "low")
        "Client completed weekly check-in reported stable mood",
        "Probationer completed all required community service hours",
        "Subject reported positive employment status and family support",
        "Case file review indicates documentation complete and uncontested",
        "Client attended all scheduled sessions and engaged cooperatively",
        "Parolee maintained employment and met all reporting requirements",
        "Defendant completed anger management program with positive evaluation",
        "Subject demonstrated consistent compliance with medication regimen",
        "Client expressed optimism about upcoming family reunification",
        "Routine supervision meeting completed no concerns noted",
    ],
    'label': [
        'high','high','high','high','high','high','high','high','high','high',
        'low','low','low','low','low','low','low','low','low','low'
    ]
}
df = pd.DataFrame(data)
print(f"Dataset: {len(df)} records, {df['label'].value_counts().to_dict()}")
```

Cell 3 — Train and evaluate:

```
# Split into training (80%) and test (20%)
X_train, X_test, y_train, y_test = train_test_split(
```

```
    df['text'], df['label'], test_size=0.2, random_state=42
)

# Convert text to numbers and train
vectorizer = TfidfVectorizer(max_features=500)
X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)

model = MultinomialNB()
model.fit(X_train_vec, y_train)

# Evaluate
predictions = model.predict(X_test_vec)
print("Confusion Matrix:")
print(confusion_matrix(y_test, predictions))
print("\nClassification Report:")
print(classification_report(y_test, predictions))
```

Cell 4 — Test with new inputs:

```
# Test with new text (edit these to match your domain)
new_texts = [
    "Subject made threatening statements during group session",
    "Client reported stable housing and regular attendance at treatment",
    "Probationer changed address without notifying supervising officer",
]
new_vec = vectorizer.transform(new_texts)
results = model.predict(new_vec)
probs = model.predict_proba(new_vec)

for text, label, prob in zip(new_texts, results, probs):
    confidence = max(prob)
    print(f"Input: {text[:60]}...")
    print(f"  Prediction: {label} (confidence: {confidence:.2f})\n")
```

4. **Screenshot the output** of Cells 3 and 4
5. Record the accuracy, precision, and recall numbers from the Classification Report
6. Use these numbers for Part 2 below

Part 2: Calculate and Understand Your Model's Metrics (30 minutes)

Step 2.1: Build Your Confusion Matrix

Using your test results from Part 1 (either Teachable Machine or Colab), fill in this confusion matrix:

```
                          Model Predicted:
                          PHISHING       LEGITIMATE
```

```
Actual:  PHISHING        TP = ___        FN = ___
         LEGITIMATE      FP = ___        TN = ___
```

(Replace "PHISHING" and "LEGITIMATE" with your class names if different.)

Example: If your model correctly identified 4 out of 5 phishing images and 3 out of 5 legitimate images:

- TP = 4 (phishing correctly called phishing)
- FN = 1 (phishing incorrectly called legitimate)
- FP = 2 (legitimate incorrectly called phishing)
- TN = 3 (legitimate correctly called legitimate)

Step 2.2: Calculate Metrics

Using your confusion matrix values, calculate:

Accuracy:

```
Accuracy = (TP + TN) / (TP + TN + FP + FN) = ___ / ___ = ___%
```

Precision (for the "threat" class):

```
Precision = TP / (TP + FP) = ___ / ___ = ___%
```

"When my model flags something as a threat, how often is it actually a threat?"

Recall (for the "threat" class):

```
Recall = TP / (TP + FN) = ___ / ___ = ___%
```

"Of all actual threats, how many did my model catch?"

F1 Score:

```
F1 = 2 x (Precision x Recall) / (Precision + Recall) = ___%
```

Step 2.3: Interpret Your Metrics

Answer these questions in your notes (you'll include them in your report):

1. **Is your precision higher or lower than your recall?** What does that mean for your model's behavior?
2. **Which errors are more costly for your domain?** False positives (false alarms) or false negatives (missed threats)?

3. **What confidence threshold would you set?** Based on the confidence scores in your test results, at what threshold would you split between "auto-action" and "human review"?
4. **What would improve your model?** More training data? More diverse data? Different classes?

> **Tip:** With only 10 test images, your metrics are rough estimates, not definitive measurements. In a real project, you'd test with 100+ examples. The goal here is understanding how to calculate and interpret the metrics.

Part 3: Find and Test Fine-Tuned Models (45 minutes)

Now shift from image classification to text classification. You'll find fine-tuned models on Hugging Face that are specifically trained for security (or your domain) and compare them against the generic models you used in Week 4.

Step 3.1: Search Hugging Face for Fine-Tuned Models

1. Go to: **https://huggingface.co/models**
2. Click **"Text Classification"** in the task filter (left sidebar)
3. Search using keywords relevant to your capstone:

Your Domain	Search Terms to Try
Cybersecurity	`phishing detection`, `malware classification`, `cyber threat`
Digital Forensics	`spam detection`, `document classification`
Fraud Investigation	`fraud detection`, `spam classification`
Intelligence Analysis	`news classification`, `topic classification`
Forensic Psychology	`emotion detection`, `sentiment fine-tuned`, `go_emotions`
Criminal Justice	`legal text`, `document classification`
Security Management	`anomaly`, `incident classification`

4. **Sort by "Most Downloads"** to find well-tested models
5. **Open the model card** for each candidate — look for:
    - Training dataset size (bigger = better)
    - Reported metrics (precision, recall, F1)
    - Recent activity (updated within last 12 months)
    - The Inference API widget on the right side (if present, you can call it from n8n)

Step 3.2: Select Two Fine-Tuned Models

Choose 2 fine-tuned models that:

- Are relevant to your capstone project
- Have the Inference API available (look for the "Inference API" widget on the model page)
- Have a model card with reported metrics

Suggested starting points (if your search is slow):

Model	What It Does	Good For
`mrm8488/bert-tiny-finetuned-sms-spam-detection`	Spam vs ham classification	Phishing/spam projects
`SamLowe/roberta-base-go_emotions`	28 emotion categories	Threat communication, psychology
`cardiffnlp/twitter-roberta-base-sentiment-latest`	3-class sentiment (neg/neu/pos)	Refined sentiment analysis
`j-hartmann/emotion-english-distilroberta-base`	7 emotion categories	Communication analysis
`papluca/xlm-roberta-base-language-detection`	Language detection	Multilingual intel analysis

Record your two selected models:

- **Fine-tuned Model A:** _____ (name + what it does)
- **Fine-tuned Model B:** _____ (name + what it does)

Step 3.3: Test Models in the Hugging Face UI

Before building the n8n workflow, test each model manually:

1. On the model page, find the **"Inference API"** widget (right side panel)
2. Type one of your test inputs into the text box
3. Click **"Compute"**
4. Record the output labels and confidence scores
5. Repeat for all 5 of your test inputs

This confirms the model works and helps you understand its output format before wiring it into n8n.

Part 4: Build the Generic vs Fine-Tuned Comparison Workflow (45 minutes)

Step 4.1: Create a New Airtable Table

1. In your existing Airtable base (or create a new base called `Week 5 Model Training`), create a table named: `Fine-Tuned Comparison`
2. Set up these fields:

Field Name	Field Type
input_text	Long text
generic_label	Single line text
generic_score	Number (4 decimal places)
finetuned_a_label	Single line text

Field Name	Field Type
finetuned_a_score	Number (4 decimal places)
finetuned_b_label	Single line text
finetuned_b_score	Number (4 decimal places)
winner	Single line text
notes	Long text
tested_at	Date (include time)

3. Add your 5 test input texts in the `input_text` field (leave all other fields blank — your workflow fills them):

For Cybersecurity (default):

1. `Unauthorized login from IP 198.51.100.4 traced to Moscow targeting user John Miller at 3:47 AM`
2. `Routine firewall rule update completed on fw-01 during scheduled maintenance window`
3. `Phishing email with spoofed Amazon domain detected targeting finance@acmecorp.com`
4. `Multiple failed SSH attempts from Beijing IP on Cloudflare production server — 47 attempts in 5 minutes`
5. `System resource utilization normal across all monitored hosts — no anomalies detected`

Or use your domain-specific equivalents from Appendix A below.

> **PDF Copy-Paste Warning:** When copying text from this PDF, long lines that wrap may include invisible line breaks. After pasting into Airtable, click into the cell and check for accidental line breaks — delete any you find. Line breaks in your input text will cause "invalid JSON" errors in the HTTP Request nodes.

> **Tip:** If you already created these records in Week 4, you can reuse the same Airtable base — just add the fine-tuned model columns to your existing table, or create a new `Fine-Tuned Comparison` table and copy the inputs over.

Step 4.2: Create a New n8n Workflow

1. In n8n, click **"New Workflow"**
2. Name it: `Week 5 - Generic vs Fine-Tuned Comparison`

Step 4.3: Add the Input Nodes

Node 1: Airtable (Search records)

1. Add an **Airtable** node
2. Set operation to **Search**
3. Connect to your `Week 5 Model Training` base (or whichever base you used), `Fine-Tuned Comparison` table
4. In the **Filter By Formula** field, enter: `{input_text} != ""`

5. This reads all records that have input text (skips blank rows)

> **Why Search instead of List?** The Search operation returns flat field access (`$json.input_text`), while List nests fields under `$json.fields.input_text`. Search also lets you filter directly with a formula so only relevant records are returned.

Node 2: Loop Over Items

1. Add a **Loop Over Items** node after Airtable (search for "Loop Over Items" or "Split In Batches" — it's the same node)
2. Set **Batch Size** to `1`

Step 4.4: Add the Generic Model (Your Week 4 Baseline)

Node 3: HTTP Request (Generic Sentiment)

1. Add an **HTTP Request** node connected to Loop Over Items
2. Configure:
   - **Method:** POST
   - **URL:** `https://router.huggingface.co/hf-inference/models/distilbert/distilbert-base-uncased-finetuned-sst-2-english`
   - **Authentication:** Header Auth → `Hugging Face API`
   - **Body Content Type:** JSON
   - **Body:**

     ```
     {
       "inputs": "{{ $json.input_text }}"
     }
     ```

3. Rename: `Generic Sentiment`
4. Under **Settings**, set **On Error** to **"Continue"** (so one error doesn't stop the whole workflow)

Node 4: Extract Generic

1. Add an **Edit Fields (Set)** node after Generic Sentiment
2. Rename it: `Extract Generic`
3. Add fields:
   - `generic_label` → `{{ $json[0].label }}`
   - `generic_score` → `{{ $json[0].score }}`
   - `input_text` → `{{ $('Search records').item.json.input_text }}`
   - `record_id` → `{{ $('Search records').item.json.id }}`

Step 4.5: Add Fine-Tuned Model A

Node 5: HTTP Request (Fine-Tuned A)

1. Add a **new HTTP Request** node — connect it to **Loop Over Items** (parallel with Generic)
2. Configure:
   - **Method:** POST

- **URL:** `https://router.huggingface.co/hf-inference/models/[YOUR-MODEL-A-NAME]` (Replace `[YOUR-MODEL-A-NAME]` with the full model name, e.g., `mrm8488/bert-tiny-finetuned-sms-spam-detection`)
- **Authentication:** Header Auth → `Hugging Face API`
- **Body Content Type:** JSON
- **Body:**

```
{
  "inputs": "{{ $('Search records').item.json.input_text }}"
}
```

3. Rename: `Fine-Tuned A`
4. Under **Settings**, set **On Error** to **"Continue"**

Node 6: Extract Fine-Tuned A

1. Add an **Edit Fields (Set)** node after Fine-Tuned A
2. The output format depends on the model. Most text classification models return:

```
[[{"label": "LABEL", "score": 0.99}, {"label": "OTHER", "score": 0.01}]]
```

3. Add fields:
   - `finetuned_a_label` → `{{ $json[0].label }}`
   - `finetuned_a_score` → `{{ $json[0].score }}`

> **Troubleshooting output format:** After adding the HTTP Request node, run it once with a test input. Double-click the node and look at the **Output** panel on the right. This shows the exact JSON structure so you can write the correct expression.

Step 4.6: Add Fine-Tuned Model B

Node 7: HTTP Request (Fine-Tuned B)

1. Add a **third HTTP Request** node — connect it to **Loop Over Items** (third parallel branch)
2. Configure the same way as Fine-Tuned A, but with your second model's URL:
   - **URL:** `https://router.huggingface.co/hf-inference/models/[YOUR-MODEL-B-NAME]`
   - All other settings are the same
3. Rename: `Fine-Tuned B`
4. Under **Settings**, set **On Error** to **"Continue"**

Node 8: Extract Fine-Tuned B

1. Add fields:
   - `finetuned_b_label` → `{{ $json[0].label }}`
   - `finetuned_b_score` → `{{ $json[0].score }}`

Step 4.7: Merge and Store Results

Node 9: Merge

1. Add a **Merge** node
2. Connect all three Extract nodes (Extract Generic, Extract Fine-Tuned A, Extract Fine-Tuned B) into it
3. Set Mode: **"Merge by Position"**

Node 10: Airtable (Update record)

1. Add an **Airtable** node after Merge
2. Set operation to **Update**
3. Connect to your base and `Fine-Tuned Comparison` table
4. Set **Record ID** to: `{{ $json.record_id }}`
5. Map the fields:
   - `generic_label` → `{{ $json.generic_label }}`
   - `generic_score` → `{{ $json.generic_score }}`
   - `finetuned_a_label` → `{{ $json.finetuned_a_label }}`
   - `finetuned_a_score` → `{{ $json.finetuned_a_score }}`
   - `finetuned_b_label` → `{{ $json.finetuned_b_label }}`
   - `finetuned_b_score` → `{{ $json.finetuned_b_score }}`
   - `tested_at` → `{{ new Date().toISOString() }}`

Step 4.8: Run and Verify

1. Click **"Test Workflow"**
2. Watch for green checkmarks on each node
3. Check your Airtable table — all 5 rows should now have results from all 3 models
4. If any node fails, double-click it to see the error output

> **Common issues:**
>
> - **Model loading (503):** The model is cold-starting. Wait 30 seconds and retry.
> - **Different output format:** Some models return `[[{...}]]` (double-nested). If `$json[0].label` doesn't work, try `$json[0][0].label` or run the node once and inspect the Output panel.
> - **Model doesn't support Inference API:** Go back to Step 3.2 and pick a different model that has the API widget on its HF page.

Part 5: Write Your Training & Evaluation Report (30 minutes)

Step 5.1: Create the Report File

Create a file called `report.md` in your text editor.

Step 5.2: Fill In the Template

```
# Week 5 Report: AutoML Training & Fine-Tuned Model Evaluation

**Name:** [Your Name]
**Date:** [Date]
**Capstone Project:** [Project name]
```

```
**My Component:** [Which component you own]

## Part A: Teachable Machine Training

### Training Setup
- **Task:** [e.g., Phishing vs Legitimate email screenshot classification]
- **Training images per class:** [number]
- **Test images per class:** [number]
- **Total training time:** [e.g., ~30 seconds]

### Test Results

| # | Actual Class | Predicted Class | Confidence | Correct? |
|---|---|---|---|---|
| 1 | | | | |
| 2 | | | | |
| 3 | | | | |
| 4 | | | | |
| 5 | | | | |
| 6 | | | | |
| 7 | | | | |
| 8 | | | | |
| 9 | | | | |
| 10 | | | | |

### Confusion Matrix

|  | Predicted: [Class 1] | Predicted: [Class 2] |
|---|---|---|
| **Actual: [Class 1]** | TP = ___ | FN = ___ |
| **Actual: [Class 2]** | FP = ___ | TN = ___ |

### Calculated Metrics
- **Accuracy:** ___%
- **Precision:** ___%
- **Recall:** ___%
- **F1 Score:** ___%

### Interpretation
[2-3 sentences: Is your model better at precision or recall? Which errors did it
make?
What would improve it?]

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## Part B: Generic vs Fine-Tuned Model Comparison

### Models Tested
1. **Generic:** distilbert-base-uncased-finetuned-sst-2-english (sentiment)
2. **Fine-Tuned A:** [model name] — [what it does]
3. **Fine-Tuned B:** [model name] — [what it does]

### Results
```

```
| Input | Generic Label (Score) | Fine-Tuned A Label (Score) | Fine-Tuned B Label
(Score) | Best Model |
|-------|----------------------|----------------------------|------------------
--------|------------|
| Record 1 | | | | |
| Record 2 | | | | |
| Record 3 | | | | |
| Record 4 | | | | |
| Record 5 | | | | |

### Analysis

**Generic model strengths:** [When did the generic model perform well?]

**Generic model weaknesses:** [When did it fail or give low confidence?]

**Fine-tuned model advantage:** [Where did the fine-tuned models clearly
outperform the generic one?]

**Biggest surprise:** [What result did you not expect?]

### Recommended Model for My Capstone Component

**Component:** [Your capstone component name]

**Primary model:** [Name] — [Why this model for your task]

**Confidence threshold:** [What threshold would you use? Why?]

**Priority metric:** [Precision, Recall, or F1? Why does this matter for your
project?]

---

## Limitations & Next Steps

[What would you do differently with more data or time? Would you fine-tune your
own
model? What additional models would you test?]
```

Part 6: Upload to GitHub (15 minutes)

Step 6.1: Set Up Your Week 5 Folder

In your GitHub repository, create this folder structure:

```
week-05-automl-training/
├── README.md
├── workflow.json
├── report.md
```

```
├── results/
│   └── comparison-table.csv
├── teachable-machine/
│   └── screenshots/
│       ├── training-interface.png
│       ├── test-result-correct.png
│       └── test-result-incorrect.png
└── metrics/
    └── confusion-matrix.md
```

Step 6.2: Export Your n8n Workflow

1. In n8n, open your `Week 5 - Generic vs Fine-Tuned Comparison` workflow
2. Click the **three dots menu** (top right) → **Download**
3. Rename to `workflow.json`

Step 6.3: Export Your Airtable Results

1. In Airtable, open the `Fine-Tuned Comparison` table
2. Click the **grid menu** (three lines) → **Download CSV**
3. Save as `results/comparison-table.csv`

Step 6.4: Create Your README

```
# Week 5: AutoML & No-Code Model Training

Trained a custom image classifier with Google Teachable Machine and compared
generic vs fine-tuned Hugging Face models for the [component name] component
of our [capstone project name].

## Custom Model Training
- Built a [Phishing/Legitimate] image classifier with Teachable Machine
- Achieved ___% accuracy on 10 held-out test images
- Precision: ___% | Recall: ___% | F1: ___%

## Fine-Tuned Model Comparison
Compared 3 models (1 generic + 2 fine-tuned) on 5 test inputs:
- Generic: distilbert-sst-2 (sentiment)
- Fine-Tuned A: [model name]
- Fine-Tuned B: [model name]

## Finding
Recommended [model name] for [component name] because [1 sentence reason].
Fine-tuned models showed [higher/comparable] performance with [more relevant
labels /
higher confidence / better handling of edge cases].

See `report.md` for full analysis.
```

Step 6.5: Upload Everything

Upload all files to your `week-05-automl-training/` folder on GitHub.

Deliverables Checklist

Before submitting, ensure you have:

- ☐ Teachable Machine classifier trained with 20+ images per class
- ☐ 10 test images evaluated (5 per class, held out from training)
- ☐ Screenshots of Teachable Machine training interface and test results
- ☐ Confusion matrix filled in with your test results
- ☐ Accuracy, precision, recall, and F1 calculated
- ☐ 2 fine-tuned Hugging Face models identified and tested
- ☐ n8n workflow comparing generic vs fine-tuned models (3 parallel API calls)
- ☐ Airtable `Fine-Tuned Comparison` table with all 5 records populated
- ☐ `comparison-table.csv` exported from Airtable
- ☐ `report.md` with:
  - ☐ Part A: Teachable Machine results, confusion matrix, calculated metrics
  - ☐ Part B: Generic vs fine-tuned comparison with analysis
  - ☐ Recommended model for capstone component with reasoning
  - ☐ Priority metric chosen (precision, recall, or F1) with justification
- ☐ All files uploaded to GitHub under `week-05-automl-training/`

Bonus Challenges (Optional)

Challenge 1: Deploy Your Teachable Machine Model to Hugging Face Spaces

Turn your Teachable Machine model into a live API that n8n can call.

1. **Export your model** from Teachable Machine as Keras (Step 1.8)
2. **Create a Hugging Face Space:**
   - Go to https://huggingface.co/new-space
   - Name: `phishing-classifier` (or similar)
   - SDK: **Gradio**
   - Click **Create Space**
3. **Upload these files** to the Space (click "Files" tab → "Add file"):

`app.py` (copy this exactly):

```
import gradio as gr
from PIL import Image
import numpy as np
from keras.models import load_model

model = load_model("keras_model.h5", compile=False)
class_names = open("labels.txt", "r").read().splitlines()
```

```
def classify(image):
    image = image.resize((224, 224))
    image_array = np.asarray(image).astype(np.float32) / 255.0
    data = np.expand_dims(image_array, axis=0)
    prediction = model.predict(data)
    confidences = {class_names[i]: float(prediction[0][i]) for i in
range(len(class_names))}
    return confidences

demo = gr.Interface(
    fn=classify,
    inputs=gr.Image(type="pil"),
    outputs=gr.Label(num_top_classes=3),
    title="Phishing Screenshot Classifier",
    description="Trained with Google Teachable Machine. Upload an email
screenshot."
)
demo.launch()
```

`requirements.txt` (copy this exactly):

```
tensorflow
Pillow
gradio
numpy
```

4. **Upload your model files:** `keras_model.h5` and `labels.txt` from the Teachable Machine export
5. Wait for the Space to build (2–5 minutes)
6. Once live, your model has an API at: `https://[username]-[space-name].hf.space/api/predict`

Challenge 2: Add Confidence-Based Routing to Your Workflow

Extend your n8n workflow with an IF node after the Merge:

- If the fine-tuned model's score > 0.85 → create a "High Confidence" record in a second Airtable table
- If score <= 0.85 → create a "Needs Review" record
- Track how many records go to each path

Challenge 3: Test with More Data

Expand your test dataset to 10 records (instead of 5). Add edge cases that you think would be hard for the models:

- Ambiguous language that could go either way
- Very short inputs (under 10 words)
- Technical jargon specific to your domain
- Inputs that combine multiple signals (e.g., "Routine maintenance on firewall after detecting suspicious outbound traffic")

Document how the models handle these harder cases in your report.

Common Issues & Solutions

"Teachable Machine won't load or train"

- Use Chrome — Teachable Machine works best in Chromium-based browsers
- Make sure you have a stable internet connection
- Clear browser cache and reload if the training button doesn't respond

"My Teachable Machine model classifies everything as one class"

- Check your training data — do you have roughly equal numbers in each class?
- Look at your images — are they diverse enough? 20 copies of the same image won't train a useful model
- Try adding more images (30+ per class) with greater variety

"I can't find fine-tuned models relevant to my domain"

- Broaden your search terms: try "text classification" without domain keywords
- Use `SamLowe/roberta-base-go_emotions` (emotion detection) or `cardiffnlp/twitter-roberta-base-sentiment-latest` (3-class sentiment) — these work across domains
- Check that you selected "Text Classification" in the task filter

"The fine-tuned model's output format is different from the generic model"

- This is expected — different models return different label names and array structures
- Run the HTTP Request node once and inspect the Output panel
- Adjust your Edit Fields expressions based on the actual output
- Common patterns:
  - Single-label models: `$json[0].label` and `$json[0].score`
  - Multi-label models: `$json[0][0].label` (extra nesting level)
  - Emotion models: may return all labels; use `$json[0].label` for the top one

"The Merge node combines results incorrectly"

- Ensure all three Edit Fields nodes connect to the same Merge node
- Set Merge mode to "Merge by Position"
- If results are misaligned, try processing one record at a time (Batch Size = 1 in Loop Over Items)

"My metrics seem wrong (e.g., 100% accuracy on 10 images)"

- Double-check that your test images were NOT included in your training set
- With only 10 test images, metrics can be unstable — getting 9/10 correct gives 90%, but getting 10/10 may mean your test images are too easy
- This is why more test data is better — see Bonus Challenge 3

Resources

- **Google Teachable Machine:** https://teachablemachine.withgoogle.com
- **Hugging Face Model Hub:** https://huggingface.co/models
- **Hugging Face AutoTrain:** https://huggingface.co/autotrain
- **Google Colab:** https://colab.research.google.com
- **Confusion Matrix Explained (visual):** https://en.wikipedia.org/wiki/Confusion_matrix
- **n8n Merge Node Docs:** https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.merge/
- **Mermaid Live Editor:** https://mermaid.live

Appendix A: Domain-Specific Test Records

Use these pre-written test inputs if you prefer domain-specific examples over the default cybersecurity ones. Each domain has 20 records — the first 10 are **high-concern** scenarios and the last 10 are **low-concern / routine**. This balanced split lets you evaluate how well generic vs fine-tuned models distinguish between actionable alerts and normal activity.

> **Tip:** You only need 5 records for the main lab. Pick a mix (e.g., 3 high-concern + 2 low-concern) so your comparison covers both extremes. Use the full 20 for Bonus Challenge 3.

Digital Forensics

High concern (1–10):

1. `Deleted file recovery initiated by analyst James Chen on NYPD workstation WS-047 at 11:52 PM — file: passwords_backup.xlsx`
2. `Registry key modification detected on Maria Torres's laptop in HKCU\Software\Microsoft\Windows\CurrentVersion\Run — timestamp: 2024-01-15`
3. `USB device connected 47 times to air-gapped terminal at Brooklyn Evidence Lab in 6-hour window — user badge not logged`
4. `System audit log shows no events between 14:00 and 18:30 — possible log tampering or system downtime`
5. `Forensic image hash mismatch detected on evidence drive EV-2024-112 submitted by Officer Daniels — chain of custody flagged`
6. `Unauthorized remote desktop session from external IP to Detective Park's workstation at Queens SVU — after hours`
7. `Encrypted container found on suspect laptop seized at JFK — volume header consistent with VeraCrypt hidden partition`
8. `Email archive PST file for witness Karen Liu modified after acquisition date — metadata timestamp inconsistent`
9. `Browser history on exhibit PC-3891 wiped using CCleaner 45 minutes before device was surrendered to NYPD`
10. `Steganography tool residue detected in AppData folder on Bronx precinct shared workstation — no authorized use on record`

Low concern (11–20): 11. `Scheduled disk imaging of evidence drive EV-2024-088 completed successfully — hash verified` 12. `Weekly backup of case file server completed — 847 files archived — no errors reported` 13. `Forensic workstation FW-12 software update applied — EnCase v23.4 patch installed by IT` 14. `Evidence intake form for case 2024-CF-0293`

processed — all fields complete — logged by Clerk Martinez 15. Chain of custody transfer for exhibit 44B signed by both parties — documentation complete 16. Monthly license audit for forensic tools completed — all seats accounted for across Manhattan labs 17. Calibration check on write-blocker unit WB-07 passed — next scheduled check in 90 days 18. Training lab environment reset after Friday workshop — all student images deleted per policy 19. File hash database updated with 12,400 new NSRL entries — routine quarterly refresh 20. Forensic report for case 2024-CF-0187 submitted to ADA Thompson — standard turnaround time

Fraud Investigation

High concern (1–10):

1. Wire transfer of $47,500 from Chase Bank account initiated from new device — destination: Lagos — account opened 3 days ago
2. Credit card belonging to David Kim used for 12 transactions in 4 countries within 6 hours — cardholder in Miami
3. Loan application submitted to Wells Fargo with same Chicago address as 8 other recent applications — different applicant names
4. Account holder reported no travel but Visa card used for $8,200 purchase at electronics store in São Paulo
5. Insurance claim filed by Margaret Chen for vehicle theft — same VIN reported stolen twice in 14 months
6. Wire recall requested by Bank of America for $125,000 transfer to shell company registered in Cyprus three days prior
7. Employee payroll direct deposit changed to offshore account 48 hours before quarterly bonus disbursement at Deloitte
8. Synthetic identity flagged — SSN issued 2019 linked to applicant claiming 15-year credit history at Capital One
9. POS terminal at QuickMart Brooklyn location processed 340 identical $49.99 transactions in single overnight shift
10. Mortgage application from James Wright lists annual income of $340,000 — employer Meridian Corp has no business registration

Low concern (11–20): 11. Monthly salary deposit of $4,200 processed — expected amount from known employer 12. ATM withdrawal of $200 at branch location — customer's home branch — 2:00 PM Tuesday 13. Recurring utility payment of $127.43 processed on schedule — same amount as previous 6 months 14. Customer updated mailing address from Manhattan to Brooklyn — verified via branch visit with photo ID 15. Annual fraud awareness training completed by 98% of branch staff — 3 pending by Friday deadline 16. Quarterly suspicious activity report filing completed on time — 14 SARs submitted to FinCEN 17. Mobile deposit of paycheck for $2,847.50 — amount consistent with prior deposits from same employer 18. Account holder called to confirm recent Amazon purchase of $89.99 — verified as legitimate 19. Merchant dispute resolved — refund of $34.50 issued to cardholder after vendor confirmation 20. Daily reconciliation of teller drawers completed — all 6 stations balanced within tolerance

Intelligence Analysis

High concern (1–10):

1. `SIGINT intercept references meeting between known extremist cell and unknown foreign nationals — location: Geneva`
2. `Source reports increased chatter from contacts in Karachi regarding infrastructure targeting — unverified — single source`
3. `Viktor Petrov photographed meeting Entity B in Istanbul for the third time this month — prior relationship unknown`
4. `Satellite imagery shows new construction at military facility near Natanz — footprint consistent with centrifuge hall expansion`
5. `Encrypted communication burst detected from previously dormant node in Hamburg — pattern matches 2019 pre-operational activity`
6. `Asset CARDINAL reports target organization acquired industrial chemicals from supplier in Ankara — quantities exceed commercial use`
7. `Social media account linked to threat actor Nightfall posted coded message matching previously identified activation phrase`
8. `Border crossing records show 4 individuals on watchlist entered Schengen zone via different entry points within 72 hours`
9. `Financial intelligence shows $2.3M routed through Liechtenstein shell companies to accounts associated with arms broker in Belgrade`
10. `UAV surveillance captured convoy of 12 unmarked vehicles departing compound near Mosul — destination unknown`

Low concern (11–20): 11. `Weekly intelligence bulletin summary — no new threat indicators identified for assessed period` 12. `Open source media confirms leadership transition in target organization — no operational impact expected` 13. `Routine liaison meeting with GCHQ counterparts completed — information sharing agreement renewed for FY2025` 14. `Publicly available trade data confirms normal shipping volume through Strait of Hormuz — consistent with seasonal patterns` 15. `Annual threat assessment draft circulated for internal review — comments due by end of month` 16. `Training exercise COBALT SHIELD completed at Fort Meade — all learning objectives met — AAR filed` 17. `OSINT collection on target media outlet shows standard editorial content — no propaganda shift detected` 18. `Diplomatic cable summarizes routine bilateral meeting between ambassadors — no classified topics discussed` 19. `Quarterly review of source reliability ratings completed — no downgrades — 3 sources upgraded to B-2` 20. `Translation backlog reduced to 12 items — all priority intercepts processed within 24-hour SLA`

Forensic Psychology

High concern (1–10):

1. `Subject made explicit threats toward former partner during intake at Bellevue Hospital — refused to provide address`
2. `Dr. Sarah Lopez noted significant agitation during structured assessment of evaluee — terminated session early`
3. `PCL-R assessment administered at Rikers Island — preliminary scoring indicates elevated psychopathy indicators`

4. Client disclosed active suicidal ideation with specific plan during group session at Bronx Community Center
5. Evaluee became physically aggressive toward staff during competency evaluation at Kings County Hospital — restraint applied
6. Risk assessment for parolee Nathan Cross indicates escalating pattern — domestic violence calls increased from 1 to 4 in 90 days
7. Forensic interview of minor at Brooklyn CAC revealed disclosure of ongoing abuse — suspect identified as household member
8. Dr. Amanda Rivera's threat assessment of inmate Hector Ruiz flags fixation on specific correctional officer — letter evidence collected
9. Neuropsychological testing of defendant shows malingering indicators — effort test scores below chance level
10. Client exhibited dissociative episode during trauma-focused session — required crisis intervention by Dr. Patel at Mount Sinai

Low concern (11–20): 11. Client completed weekly check-in call — reported stable mood and consistent medication compliance 12. Subject reported positive employment status and ongoing family support system at 6-month follow-up 13. Group therapy session at Queensboro Correctional completed — 8 of 10 participants engaged appropriately 14. Routine competency restoration progress note filed — defendant attending all classes and responding to medication 15. Annual recertification training on forensic interviewing protocols completed by all staff at Manhattan CAC 16. Client maintained sobriety for 120 consecutive days — confirmed by weekly urinalysis and sponsor report 17. Parenting evaluation for family court completed — both parties cooperative — report submitted to Judge Williams 18. Psychoeducational assessment for juvenile respondent completed — results consistent with age-appropriate functioning 19. Treatment plan review for client at Harlem outpatient clinic — all goals on track — next review in 90 days 20. Intern supervision notes reviewed and signed — no clinical concerns identified — training progressing normally

Criminal Justice

High concern (1–10):

1. Defendant Robert Garcia failed to appear at Manhattan Criminal Court for third scheduled date — warrant issued
2. Parolee Michael Johnson reported new address in Bronx without required advance notification to Officer Davis
3. Sentencing recommendation submitted by prosecution — requesting upward departure from standard guidelines
4. Inmate altercation at Rikers Island resulted in hospitalization of two individuals — weapon recovered from cell block C
5. GPS ankle monitor for probationer Lisa Grant shows presence at victim's workplace on three occasions this week
6. Defendant's pretrial release revoked after arrest for new offense — original charges include assault in the second degree
7. Victim advocate reports witness intimidation complaint filed against associate of defendant Marcus Wells

8. `Probation officer Torres documents third positive drug test for parolee — mandatory revocation hearing scheduled`
9. `Juvenile respondent Anthony Davis absconded from residential placement in Westchester — missing 72 hours`
10. `Body-worn camera footage contradicts officer testimony in suppression hearing — defense moves to dismiss at Brooklyn Supreme Court`

Low concern (11–20): 11. `Probationer completed all required community service hours — supervisor report positive` 12. `Case file review indicates chain of custody documentation complete and uncontested` 13. `Defendant completed anger management program with positive evaluation — certificate filed with court` 14. `Monthly restitution payment of $150 received on schedule — balance remaining: $2,400` 15. `Routine supervision meeting with probationer — employed full-time, stable housing, no new contacts with law enforcement` 16. `Court interpreter request for Mandarin filed for next hearing date — confirmed available` 17. `Expungement petition for 2018 misdemeanor reviewed — applicant meets all eligibility criteria` 18. `Jury selection for trial 2024-CR-4471 completed — 12 jurors and 2 alternates seated without issue` 19. `Pre-sentence investigation report for non-violent first offender completed — submitted to Judge Park on schedule` 20. `Community court session processed 14 low-level cases — all resolved with services referrals — no contested matters`

Security Management

High concern (1–10):

1. `Unauthorized access attempt by Apex Security contractor at Northfield HQ server room 3B — 2:17 AM`
2. `Visitor badge issued to Pinnacle Consulting representative — did not return badge at end of shift — facility unsecured`
3. `Tailgating incident captured on CCTV at Goldman Sachs Tower main entrance — second occurrence this week`
4. `Security officer at Brookfield Place reports individual photographing access control panels and emergency exits`
5. `Master key card for floors 14–22 at MetLife Building reported lost by cleaning contractor Prestige Services`
6. `Perimeter fence sensor triggered 3 times overnight at ConEdison substation — no corresponding CCTV footage available`
7. `Former employee badge of terminated analyst Sarah Kim used at JPMorgan Chase data center — badge should have been deactivated`
8. `Fire exit door at Amazon warehouse JFK8 propped open during overnight shift — alarm was manually silenced`
9. `Guard tour system shows Officer Ramirez skipped 4 of 12 checkpoints during overnight patrol at Pfizer facility`
10. `Unregistered drone detected within restricted airspace above Citigroup headquarters — FAA notification pending`

Low concern (11–20): 11. `Monthly fire suppression system test completed — all zones passed — logged by facilities team` 12. `All access control panels show normal status — no alarms`

active — end of shift report 13. Quarterly security awareness training completed by 94% of staff at Morgan Stanley — remaining 6% scheduled next week 14. Visitor management system software updated to version 4.2 — no downtime during installation 15. Annual penetration test of physical security completed by SecureWorks — report delivered to CISO 16. New CCTV cameras installed at parking garage levels B2 and B3 — coverage gap from 2023 audit resolved 17. Guard shift change at 0600 completed on time — all posts staffed — incident log reviewed and signed 18. Emergency evacuation drill completed in 4 minutes 12 seconds — within target time — all floors accounted for 19. Loading dock access log reconciled with delivery manifests — all 23 shipments matched — no discrepancies 20. Badge printer maintenance completed — 200 blank cards restocked — inventory log updated by Admin Lee